Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Ultrashort pulse laser Technology for scalable and wear-free ceramic processing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (X), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Mechanical & Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	Laser Processing Machines	Precision Manufacturing Machinery	Optical Energy-based Processing Machines
Concept of Proposed Technology	Pilot system for wear-free ultrashort pulse laser processing (e.g. cutting) of technical ceramics (e.g. Al_2O_3, ZrO_2, Si_3N_4) for semiconductor applications, high power electrical applications and cutting tools in industrial environment at Korean SME, including special optical systems for increased productivity as fast scanners and multi beam optical systems • ⬭ Demonstration of functional part manufacturing			
Rationale for Joint R&D Project	- Quality of processed part and productivity of ultrashort pulse laser process for many applications are not meeting the requirements - Optical systems for large parts and high productivity must still be developed and industrialized - Functionality of laser-processed part must be investigated			

Key Words	1. Processing of ceramics 2. Ultrashort pulse laser 3. High-power electronics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)

R&D Objectives	- Development of system technology for process scaling
	  ○ Fast beam deflection systems
	  ○ Rapid part movement
	  ○ Multi beam system
	- Development of a highly productive ultrashort pulse cutting process for ceramics
	- Evaluation of technology platforms for series production			

R&D Target Specifications	Processing of different technical ceramics (Al_2O_3, ZrO_2, Si_3N_4) with the following parameters - ablation rate >20 mm³/min - depth accuracy ±100 nm - damage (cracks, chipping) below detection limit			
R&D Plan	○ Process development for crack-free laser ablation of ceramics using a wavelength of 1030 nm and a pulse duration of 1 ps ○ Productivity scaling through process and system adaptation ○ Fabrication of application-relevant ceramic demonstrator parts ○ Optical and mechanical design of the pilot system ○ Process validation and technology transfer to Korean partner			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer ILT will act as the key technology provider and process developer. Using its existing high-end infrastructure (including high-power ultrashort pulse lasers, high-speed galvanometer and polygon scanners), Fraunhofer ILT will develop and validate the laser ablation process of ceramics in Germany. Based on this groundwork, a pilot system will be designed and transferred to the Korean SME for procurement, assembly and industrial testing. Expected contributions and outputs: ⬭Development of process parameters and scaling strategies using Fraunhofer's existing equipment ⬭Fabrication and evaluation of functional demonstrator parts ⬭System specification, component selection, and setup planning ⬭Provision of processing guidelines, parameter sets, training materials and hands-on training at the Korean site ⬭Support in IP development and protection
Expected R&R of Korean SME Partner	The Korean SME will lead the consortium and take responsibility for the industrial implementation of the developed technology. Based on the pilot system design and process know-how provided by Fraunhofer, the SME will procure required components, assemble the system locally, and replicate the validated process under near-production conditions. It will also drive the commercialization and support technology localization in Korea. Expected contributions and outputs: ⬭Project coordination ⬭Provision of ceramic components and part specifications ⬭Setup of a test environment to demonstrate the laser process ⬭Procurement of laser source, scanner optics, and motion components ⬭Assembly and setup of the pilot system based on Fraunhofer's design ⬭Process replication and functional part manufacturing in a near-industrial environment ⬭Plan for commercialization and exploitation

5. Expected Outcomes/Impact

Expected Outcomes and Impact	○Pilot-level laser system for ultrafast and crack-free micromachining of ceramics, enabling Korean SME to enter high-value markets such as electronics and energy ○Elimination of wasteful and time-consuming conventional processes like mechanical grinding or EDM and, thus, less costs per part ○Generation of new IP (e.g. process design, system layout) ○Joint dissemination via conference presentations and publications

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization